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# Analysis of Temperature-Dependent Characteristics of Cr/P-Doped Si Schottky Barrier Diodes before and after 5.4 MeV $^{241}\text{Am}$ Irradiation

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**Received:** 19 August 2025 | **Revised:** 15 December 2025 | **Accepted:** 24 December 2025

**Keywords:** alpha-particle irradiation | chromium | electron beam deposition | Richardson constant | Schottky barrier diode | silicon

## ABSTRACT

Schottky barrier diodes (SBDs) based on chromium contacts on n-type silicon (Cr/n-Si) were fabricated using electron beam deposition and systematically characterised before and after 5.4 MeV alpha-particles from a  $^{241}\text{Am}$  source at a fluence of  $1.3 \times 10^{10} \text{ cm}^{-2}$ . Temperature-dependent current–voltage (*IV*) and capacitance–voltage (*CV*) measurements from 60 to 300 K revealed strong temperature and irradiation influence on Cr/n-Si SBD parameters. The reverse leakage current decreased at lower temperatures, indicating suppressed activity of deep-level defects. Alpha-particle irradiation introduced barrier height inhomogeneities and caused a slight deviation in the modified Richardson constant from the theoretical value for n-Si ( $112 \text{ A}\cdot\text{cm}^{-2}\cdot\text{K}^{-2}$ ). Deep level transient spectroscopy (DLTS) confirmed the introduction of additional electrically active defects after irradiation. Despite these changes, the diodes maintained stable rectifying behaviour, demonstrating their potential of Cr/n-Si SBDs for use in radiation-hardened and cryogenic electronic applications.

## 1 | Introduction

Transition metals such as Ti, Au, Cr, Ni, Co, Au, Pt, W and Pd are of significant interest for semiconductor device applications due to their favourable electrical, mechanical, magnetic, optical, and thermal properties [1–5]. Among these, titanium has been extensively investigated for Schottky barrier diode (SBD) owing to its thermal stability, mechanical strength, and low contact resistance [6–8]. In contrast, chromium, despite possessing a relatively high work function ( $\sim 4.50 \text{ eV}$ ), strong adhesion, and useful magnetic properties, remains comparatively underexplored for Schottky contact [9–11]. Chromium is widely used in integrated circuits, beam splitters, photomasks, and optical recording media [11–13], suggesting its potential for reliable Schottky interfaces, yet systematic studies on Cr-based SBDs, especially under irradiation and cryogenic conditions, are still limited.

The current flow across metal–semiconductor (M–S) SBDs has been widely studied both theoretically and experimentally [14, 15]. An SBD, with its low forward voltage and high-speed switching, finds use in rectification and signal processing [16, 17]. Given its characteristics, chromium is a promising material for SBDs [6, 9, 11–13, 16, 18–20]. In addition to the metal contacts, the substrate material is crucial to device performance [21, 22]. Semiconductors like Ge [23, 24], GaN [25], GaAs [5], Si [26], ZnO [27], and SiC [28] have all been explored; however, silicon, with its indirect bandgap of 1.12 eV at 273 K [29], remains the most widely used because of its prevalence in microelectronics [30, 31].

Deposition techniques also significantly influence device performance [1, 3, 23, 32]. Methods such as magnetron sputtering [32], electron beam deposition (EBD) [3], and electron beam exposure

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[22, 23] have been reported to induce electrically-active deep-level defects in quantifiable amounts at and near to M–S junctions materials like Si [21], GaN [33], Ge [23], GaAs [34], 4H–SiC [22] and 6H–SiC [35], whereas thermal resistive evaporation technique does not [36]. EBD, however, allows deposition of high melting-point metals like Cr (melting point of 1907°C) [37] and provides a faster deposition rate than magnetron sputtering deposition [38, 39].

Characterising the current–voltage (*IV*) and capacitance–voltage (*CV*) behaviour of SBDs over a range of temperatures provides insights beyond what room-temperature measurements offer [28, 40–43]. From the past studies, it has been observed that the *IV* and *CV* characteristics of the M–S devices deviate from the ideal thermionic emission (TE) theory [28, 40–44]. Therefore, the *IV* and *CV* characteristics at room temperature alone cannot suffice to characterise the diodes, and thus, there is a need to carry out the electrical characterisation of Schottky diodes at various temperatures. For example, Durmus et al. [45] have characterised electrical properties of Re/n-Si contacts using the *IV* and *CV* in the temperature range of 60–300 K. They fabricated the contact using pulsed laser deposition and concluded that the values of Schottky barrier height and ideality factor are strongly dependent on temperature. Demircioglu et al. [46] deposited Cr on n-type Si using the electrodeposition technique and carried out the characterisation at a temperature range of between 80 and 320 K. These authors used a Gaussian distribution of barrier height to discuss the abnormal temperature dependence of the Schottky barrier height as well as the ideality factor. Evans-Freeman et al. [47] thermally deposited Au Schottky contacts on Si<sup>+</sup> ions implanted in the n-Si and then carried out *IV* and *CV* measurements at the temperature range 40 to 300 K to determine the integrity of the diode as well as to verify that the depletion region could be traversed through the implanted region. They also reported the presence of three deep levels in the Au/n-Si devices by using deep level transient spectroscopy (DLTS). Also, Zhu et al. [48] reported that the electrical parameters of the neutron-irradiated GaN SBDs depend on temperature. They observed new defects, and the temperature-dependent results testify that GaN Schottky contact has high resistance to neutron irradiation. Furthermore, Omotoso et al. [49] have also reported irradiation effects on the electrical properties of Ni/4H–SiC SBDs using high-energy protons, high-energy electrons [28], and alpha-particles [40]. A great deviation in the modified Richardson constant after irradiation of the contacts, as compared to the as-deposited contacts, was reported from their findings. However, to the best of our knowledge, the electrical characteristics of alpha-particle irradiation on electron beam deposited Cr SBDs on n-Si have not been reported.

Despite the extensive literature on Si-based SBDs and their response to radiation, several gaps remain. Specifically, the temperature-dependent electrical behaviour of Cr/n-Si diodes fabricated by EBD under alpha-particle irradiation has not been investigated. Moreover, the nature of deep-level defects introduced by alpha irradiation in this material system, and their correlation with Schottky barrier inhomogeneity and morphological changes, remains unreported. This study aims to address these gaps by providing a detailed temperature-dependent *IV*, *CV*, and DLTS analysis of Cr/n-Si SBDs before and after 5.4 MeV alpha irradiation, supported by morphological characterisation. The work offers new insights into the radiation-induced degradation mechanisms

in Cr/Si interfaces and demonstrates the feasibility of Cr-based contacts for radiation-hardened low-temperature applications.

In this work, temperature-dependent *IV* and *CV* characteristics of Cr SBDs fabricated through EBD on phosphorus-doped Si substrates over the temperature range 60–300 K were studied. The choice of Cr as a beneficial material to employ for Schottky barrier contacts on silicon is, therefore, of particular interest because of its work function (4.37 eV), which is lower than that of Ni, Au and Pt, as well its good adhesion and mechanical stability [6, 50]. Considering the work function of Cr ( $\phi_{Cr} = 4.37$  eV) and the electron affinity of Si ( $\chi_{Si} = 4.05$  eV), the Schottky–Mott relation predicts a theoretical barrier height for Cr/n-type Si contacts of  $q\phi_{bn} = (q\phi_{Cr} - \chi_{Si}) = 0.32$  eV [49], sufficiently large for practical device applications.

## 2 | Experimental

In this study, n-type, phosphorus-doped epitaxially grown silicon with a thickness of between 13 and 17  $\mu\text{m}$ , resistivity of between 1.4 and 1.8  $\Omega\cdot\text{cm}$ , and a doping concentration of  $3.6 \times 10^{15} \text{ cm}^{-3}$  grown on n<sup>++</sup> Si substrate with (111) orientation was used. Samples of about  $3 \times 5 \text{ mm}^2$  were cut from the wafer and ultrasonically cleaned in trichloroethylene, isopropanol, and methanol (5 mins each) to remove grease and dust. Insoluble organic and metallic contaminants were removed using a standard SC-1 solution ( $\text{H}_2\text{O}:\text{30\% H}_2\text{O}_2:25\%\text{NH}_4\text{OH} = 5:1:1$ ) at about 80°C for 10 min, followed by a de-ionised water rinse and 40% hydrofluoric acid dip for 1 min to strip away oxide layers. Nitrogen gas was used to dry the samples before loading into a resistive vacuum chamber for ohmic contact deposition.

Ohmic contacts were formed on the unpolished side using resistively deposited gold–antimony (Au–Sb, which contained 6% of Sb) to a thickness of 2500 Å at a rate of about  $0.2 \text{ Å}\cdot\text{s}^{-1}$  under  $1.5 \times 10^{-6}$  mbar vacuum. These contacts were annealed at 370°C in argon for 10 min to improve contact quality and to avoid oxidation. The sample was only removed after the furnace cooled below 27°C. Cleaning steps were repeated after the annealing, but the HF dip was limited to 30 s to avoid delamination of the Au–Sb ohmic contact layers.

Two sets of chromium (purity of 99.95%) contacts were deposited by EBD on the epitaxially grown (polished) Si surface. The first set involved patterned deposition through a contact mask, forming circular Cr SBDs (diameter of 0.55 mm, thickness 1000 Å) and deposited at about  $0.1 \text{ Å}\cdot\text{s}^{-1}$  in a vacuum of  $1 \times 10^{-7}$  mbar. The second set, deposited without a mask, was used for morphological studies of the thin films on the n-Si substrate.

Initial *IV* and *CV* characterisation of the Cr/n-Si SBDs was performed at room temperature (27°C) in the dark using an HP 4140 B pA Metre/DC voltage source and an HP 4192A LF impedance analyser to verify diode rectification ( $n = 1.04$ ,  $\phi_{IV} = 0.54$  eV and  $\phi_{CV} = 0.67$  eV), which is consistent with the literature [51–53]. The Cr/n-Si device was mounted in a closed-cycle helium cryostat and characterised across 60–300 K in 20 K steps. The *IV* measurements were carried out over a voltage range from 1.0 (forward bias) to –2.0 V (reverse bias). The *CV* characterisation was likewise measured from 0.2 (forward bias) to –2.0 V (reverse bias). Thereafter, DLTS measurements were conducted from 24 to 300 K at a scan rate of  $1 \text{ K}\cdot\text{min}^{-1}$  and emission rate window of

$2.5\text{--}1000\text{ s}^{-1}$  using a Boonton 7200 capacitance metre (100 mV, 1 MHz) to determine the defect signatures (apparent capture cross section,  $\sigma_n$ , and activation energy,  $E_n$ ) observed in this study [3].

After initial characterisation, the same Cr/n-Si SBDs were irradiated with 5.4 MeV  $\alpha$ -particles from a  $^{241}\text{Am}$  source at a fluence of  $1.3 \times 10^{10}\text{ cm}^{-2}$  from the flux of  $7.2 \times 10^6\text{ cm}^{-2}\text{ s}^{-1}$  at room temperature. Post-irradiation  $IV$  and  $CV$  measurements were repeated under identical conditions to assess device stability and radiation response. The  $IV$ - $T$  results enabled the evaluation of current transport mechanisms, temperature-dependent ideality factors, and barrier inhomogeneity. A Gaussian distribution model was used to extract the mean Schottky barrier height, capturing interfacial inhomogeneity effects. Post-irradiation DLTS was performed to characterise the deep-level defects introduced by alpha-particle irradiation exposure.

The second sample set was used for surface morphology analysis using a Zeiss Ultra Plus 55 FESEM (Carl Zeiss, Oberkochen, Germany) operated at an accelerating voltage of 2.0 kV to observe any changes due to irradiation.

### 3 | Results and Discussion

#### 3.1 | Capacitance–Voltage as a Function of Temperature ( $CV$ - $T$ )

$CV$  measurements, being non-destructive, provide valuable insights into the rectifying nature of the Schottky barrier contact [46]. Figure 1 shows  $C^{-2}$  as a function of reverse bias voltage,  $V$ , characteristics for the Cr/n-Si SBDs before and after alpha-particle irradiation, measured at a frequency of 1 MHz over a temperature range of 60 to 300 K. Figure 1a shows the plots of the  $C^{-2}$ - $V$  characteristics measured from 0.2 (forward bias) to  $-2.0$  V (reverse bias). The high-frequency measurement at 1 MHz was specifically selected to suppress the influence of interface state charges, as their response time is too slow to be observed at these frequencies [47, 54].

Only the plots at 60 and 300 K are shown in Figure 1, as the intermediate temperature data (80 to 280 K) exhibited negligible variation and followed a similar trend. Notably, a crossover was observed in the 60 K plots before and after irradiation. This

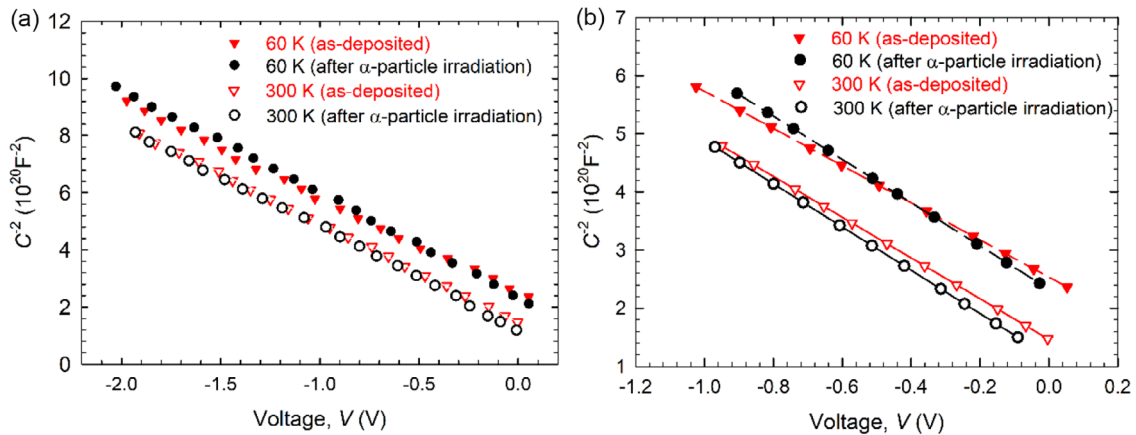
crossing may be attributed to a competing influence between thermal carrier freeze-out in the as-deposited diode and trap-assisted carrier compensation introduced by irradiation [55]. At very low temperatures, carrier freeze-out reduces free electron availability in the unirradiated sample, leading to a wider depletion region and reduced capacitance [55]. In contrast, after irradiation, the sample benefits from donor-like deep-level defects that partially compensate for freeze-out, resulting in a higher capacitance at stronger reverse bias, though a crossover occurs near zero bias which may be due field-assisted ionisation effects.

Across all temperatures, the linearity of  $C^{-2}$ - $V$  plots, Figure 1b, confirm the uniformity of net donor concentration in the depletion region and the absence of Cr-Si interfacial reactions. In all cases, irradiation decreased the intercept on the voltage axis, indicating a decrease in the  $CV$  barrier height. The gradient of the  $CV$  curve becomes less after irradiation, indicating an increase in donor concentration [56]. It may, therefore, be deduced that irradiation induces defects with a donor-like character. This effect is more pronounced at lower temperatures (60 K) than at room temperature. The capacitance was also observed to decrease with decreasing temperature, with a corresponding increase in the intercept on the voltage axis, which is consistent with classical SBD behaviour showing an increasing barrier height with decreasing temperature [57].

The extracted electrical parameters from the  $CV$  data, such as SBH, net carrier concentration, and built-in voltage, were analysed using the standard depletion capacitance equation for the SBD [28, 40, 53, 58, 59]

$$C^{-2} = 2 \left( V_{bi} + V - \frac{kT}{q} \right) / (q\epsilon_s A^2 N_T) \quad (1)$$

where  $C$  is the capacitance of the diode,  $V_{bi}$  is the built-in voltage, which is obtained as the intercept on the applied voltage axis after the extrapolation (i.e., the voltage when  $\frac{1}{C^2} = 0$ ),  $V$  is the applied reverse bias,  $k$  is the Boltzmann constant ( $8.625 \times 10^{-5}\text{ eV.K}^{-1}$ ),  $T$  is the temperature,  $q$  is the elementary charge,  $\epsilon_s$  is the semiconductor's permittivity,  $A$  is the SBD's effective area, and  $N_T$  is the net carrier concentration, which is calculated from the gradient of the linear part of  $C^{-2}$  as a function of  $V$  plot.



**FIGURE 1** | Experimental  $1/C^2$  -  $V$  characteristics of the Cr/n-Si Schottky barrier diodes (SBDs) at 60 and 300 K before and after irradiation by alpha-particles at a fluence of  $1.3 \times 10^{10}\text{ cm}^{-2}$ . (a) Shows the full voltage measurement range from  $-2.0$  to  $0.2$  V, while (b) shows the linear fitted region from (a), used to extract the electrical parameters listed in Table 1.

During reverse bias measurements, the Au–Sb ohmic contact was positively biased relative to the Cr SBD. The Schottky barrier height from CV,  $\phi_B^{CV}$ , is can then be calculated using

$$\phi_B^{CV} = V_{bi} + V_n \quad (2)$$

where  $V_n$  is the potential difference between the Fermi level ( $E_F$ ) and the conduction band minimum ( $E_C$ ) for n-Si in the neutral region. The  $V_n$  can only be computed if the concentration of the doping is known and, for the case where acceptors and defects can be neglected, is expressed as Equation (3)

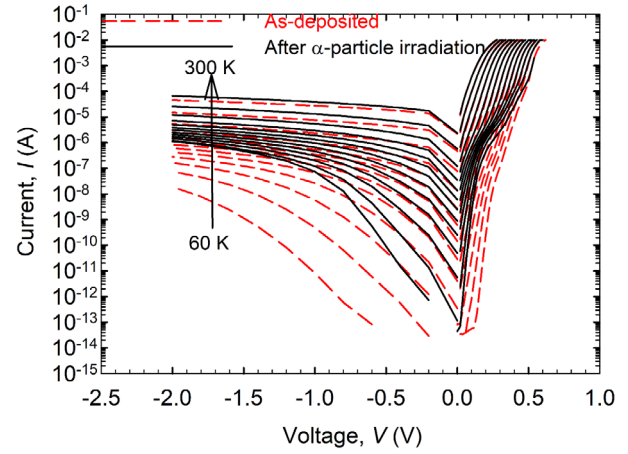
$$V_n = \frac{E_C - E_F}{q} = \frac{kT}{q} \ln \left( \frac{N_C}{N_D} \right) \quad (3)$$

where  $N_C$  is the effective density of states in the conduction band. The values obtained from the linear fit of Figure 1b are tabulated in Table 1.

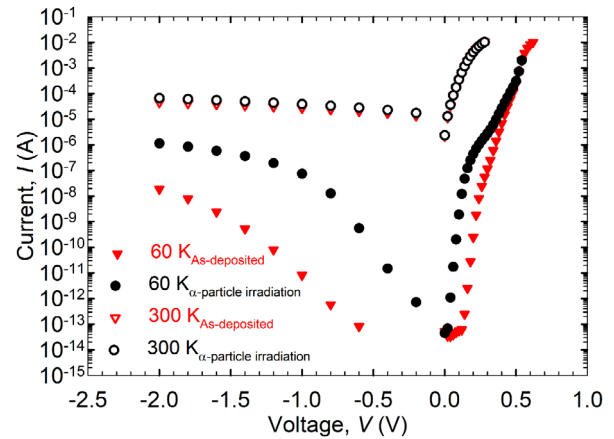
### 3.2 | Current–Voltage as a Function of Temperature (IV–T)

The IV results obtained at 300 K (as tabulated in Table 1) confirm rectifying behaviour of the SBDs both before and after irradiation, supporting their suitability for this study. Figure 2 shows the typical semi-log IV characteristics under forward and reverse bias, measured between 60 and 300 K in steps of 20 K. The IV characteristics varied gradually with temperature. In an ideal case, reverse current in a Schottky diode should be independent of reverse bias, but an increase is often observed with bias due to non-ideality.

For a better understanding, Figure 3 shows representative IV curves at 60 and 300 K before and after irradiation. The reverse bias characteristics of the plots reveal significant leakage current before and after irradiation, which increased slightly at 300 K after irradiation, as tabulated in Table 2. This radiation-induced increase in reverse leakage current became more pronounced at lower temperatures. Such behaviour may be attributed to the generation-recombination currents associated with deep-level defects induced by the irradiation [53]. At low temperatures, TE is strongly suppressed, reducing the contribution of carriers that can surmount the barrier. As a result, the leakage current is dominated by defect-assisted processes, making the effect of irradiation-induced traps move visible in the reverse bias characteristics. The forward IV plots in Figure 2 show near-linear behaviour semi-log scale at low bias, indicating TE dominance.



**FIGURE 2** | The semi-logarithmic current–voltage characteristics of the Cr/n-Si SBDs before and after irradiation by alpha-particles measured in temperature range of 60 to 300 K in 20 K steps.



**FIGURE 3** | The semi-logarithmic current–voltage characteristics of the Cr/n-Si SBDs for temperatures 60 and 300 K before and after irradiation by alpha-particles at a fluence of  $1.3 \times 10^{10} \text{ cm}^{-2}$ .

Deviation from linearity at higher bias is attributed to series resistance, while low-bias deviations, especially at lower temperatures, suggest additional transport mechanisms such as generation-recombination [53]. Since the magnitude of the deviation is roughly the same in both forward and reverse directions, it is likely that the same mechanism is involved in both. These

**TABLE 1** | Extracted electrical parameters of Cr/n-Si Schottky barrier diodes before and after room-temperature alpha-particle irradiation at a fluence of  $1.3 \times 10^{10} \text{ cm}^{-2}$  calculated from experimental IV and CV plots at 60, 200 and 300 K.

| Sample               | $n$  | $I_s$ , A              | $R_s$ , $\Omega$ | $N_T$ ,<br>$\times 10^{15} \text{ cm}^{-3}$ | $V_{bi}$ , V | $\phi_{IV}$ , eV | $\phi_{CV}$ , eV |
|----------------------|------|------------------------|------------------|---------------------------------------------|--------------|------------------|------------------|
| Unirradiated (300 K) | 1.04 | $7.22 \times 10^{-6}$  | 9                | 4.23                                        | 0.39         | 0.54             | 0.67             |
| Unirradiated (200 K) | 1.35 | $1.50 \times 10^{-9}$  | 5                | 4.95                                        | 0.55         | 0.50             | 0.55             |
| Unirradiated (60 K)  | 3.97 | $2.02 \times 10^{-14}$ | 4                | 4.64                                        | 0.79         | 0.19             | 0.83             |
| Irradiated (300 K)   | 1.05 | $7.95 \times 10^{-6}$  | 8                | 3.98                                        | 0.29         | 0.54             | 0.57             |
| Irradiated (200 K)   | 1.67 | $1.81 \times 10^{-8}$  | 4                | 4.05                                        | 0.46         | 0.45             | 0.46             |
| Irradiated (60 K)    | 4.09 | $1.63 \times 10^{-10}$ | 4                | 3.98                                        | 0.62         | 0.15             | 0.66             |



**TABLE 2** | The variation of leakage current characteristics of Cr/n-Si Schottky contacts with temperature bias before and after irradiation with alpha-particles at a fluence of  $1.3 \times 10^{10} \text{ cm}^{-2}$  measured at a reverse bias of 2.0 V.

| States                             | Temperature, K | Leakage current, A    |
|------------------------------------|----------------|-----------------------|
| Before irradiation                 | 300            | $4.62 \times 10^{-5}$ |
| After alpha irradiation for 30 min | 300            | $6.71 \times 10^{-5}$ |
| Before irradiation                 | 60             | $1.89 \times 10^{-8}$ |
| After alpha irradiation for 30 min | 60             | $1.15 \times 10^{-6}$ |

deviations were more noticeable after irradiation, which may be attributed to the presence of deep-level traps contributing to the carrier transport.

To evaluate the deviation from the ideal diode behaviour, the  $IV$  parameters such as ideality factor ( $n$ ), zero-bias Schottky barrier height ( $\phi_B^{IV}$ ), Saturation current ( $I_s$ ), and series resistance ( $R_s$ ), before and after irradiation, need to be calculated from the plots. The experimental  $IV$  characteristics shown in Figure 2 can be analysed by using the TE model given in Equation (4) [53, 58]

$$I = I_s \exp \left[ \frac{q(V - IR_s)}{nkT} \right] \left( 1 - \exp \left[ -\frac{q(V - IR_s)}{kT} \right] \right) \quad (4)$$

The saturation current,  $I_s$ , can be derived from the straight-line intercept of  $\ln I$  when  $V = 0$ , and can be written as:

$$I_s = AA^* T^2 \exp \left( -\frac{q\Phi_B^{IV}}{kT} \right) \quad (5)$$

where  $q$  is the electron charge,  $V$  is the forward bias voltage,  $IR_s$  is the voltage drop across the series resistance,  $n$  is the ideality factor which expresses the conformity of the diode to the pure TE of an ideal SBD ( $n = 1$  for an ideal diode),  $k$  is the Boltzmann constant,  $A$  is the effective area of the Cr/n-Si SBD ( $2.8 \times 10^{-3} \text{ cm}^2$ ),  $R$  is the theoretical Richardson constant of  $n$ -type Si ( $112 \text{ cm}^{-2} \cdot \text{K}^{-2}$ ) [53, 58, 60],  $T$  is the absolute temperature, and  $q\Phi_{ap}$  is the experimental zero-bias barrier height, which can be determined from Equation (5) if  $I_s$  is known. The ideality factor can be determined from the slope of the linear region of the forward bias  $IV$  characteristic as

$$n = \frac{q}{kT} \frac{dV}{d(\ln I)} \quad (6)$$

The experimental values of the electrical  $IV$  parameters were determined from Figure 2 and tabulated in Table 1. As can be seen from Table 1, the saturation current decreased with a decrease in temperature and further decreased after irradiation.

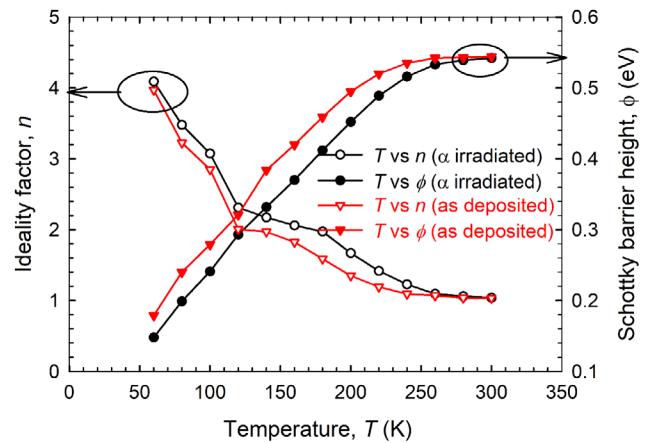
The temperature dependence of  $\phi_B^{CV}$  and  $\phi_B^{IV}$  obtained from the reverse bias  $\frac{1}{C^2} - V$  and forward bias of the  $IV$  characteristics, respectively, are tabulated in Table 1. It is worth noting that their experimental values do not agree. The disagreement in their measurements can be explained by barrier height inhomogeneities at the structure as a result of the interfacial layer or deep level defects in the  $n$ -Si [61] that affect the measuring techniques differently. The CV method averages the entire area of the Schottky and tends

to measure an average barrier height [43, 45], whereas the direct current,  $I$ , across the interface Cr/n-Si, as written in Equation (4), depends exponentially on the SBH and therefore the current is especially sensitive to small regions of low barrier height, which affect the measured barrier height disproportionality at the junction [43, 45]. Because of the aforementioned reasons, the experimentally measured  $IV$  barrier height is lower than the actual value of the weighted average of barrier heights. Therefore, the experimentally measured CV barrier height is very close to the weighted arithmetic average of the barrier height [43, 45].

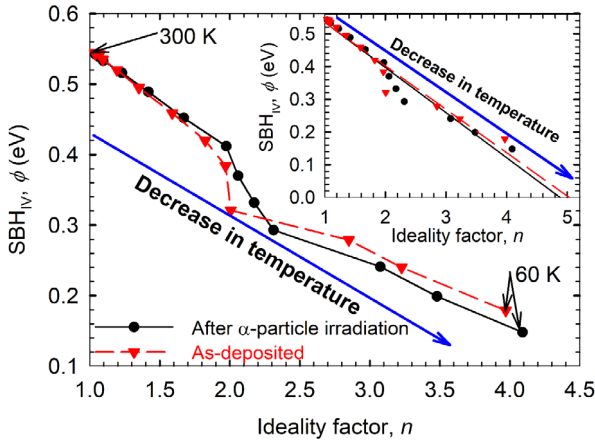
Figure 4 shows the strong temperature dependence of both  $n$  and  $\phi_B^{IV}$  of the diodes. The ideality factor increases with decreasing in temperature. The lower the temperature, the more the diode deviates from the ideal behaviour, and the deviation was slightly more pronounced after irradiation. Similarly,  $\phi_B^{IV}$  decreases with temperature and is higher for irradiated diodes. Further deviation of  $n$  and  $\phi_B^{IV}$  after irradiation may be attributed to the presence of deep-level defects causing generation-recombination current and interface states [62]. Their behaviours with temperature and irradiation could also be attributed to the lateral inhomogeneities at the interface of Cr and  $n$ -Si caused by defects, surface treatment, metal, deposition process and inhomogeneity in carrier concentration, as well as non-uniformity of the interface charges. It may also result to deviation from TE, which becomes more pronounced at lower temperatures [59, 63, 64]. At low temperature, the electrons acquire a very weak kinetic energy,  $kT$ , and their preferred path to pass is lower barriers, which will cause the current to flow through patches of the lower SBH and therefore lead to a higher value of the ideality factor [42, 46]. The variation in ideality factor and SBH with temperature and irradiation cannot be fully or correctly discussed using the ideal TE theory. The increase in ideality factor and decrease in SBH with decreasing temperature will be analysed and further examined using TE with a Gaussian distribution of the SBHs in the next section.

### 3.3 | The Influence of Inhomogeneities in the Schottky Barrier Height

Figure 5 shows the relationship between Schottky barrier height for  $IV$  ( $SBH_{IV}$ ) and ideality factor for Cr/Si SBDs,



**FIGURE 4** | Ideality factor and  $IV$  Schottky barrier height of the Cr/n-Si SBD at temperature range 60 to 300 K measured before and alpha-particle irradiation.



**FIGURE 5** | Comparison of zero-bias Schottky barrier height as function of ideality factor with temperature for the Cr/n-Si Schottky barrier contact before and after alpha-particle irradiation measured at different temperatures. The inset shows the corresponding linear fit based on Tung's barrier inhomogeneity model.

measured at different temperatures before and after irradiation. It can be seen from the plot that the  $SBH_{IV}$  decreases as the ideality factor increases. That is, the  $SBH_{IV}$  became smaller as the ideality factor increased. This inverse correlation is explained using Tung's theoretical approach [65], which attributes such behaviour to lateral inhomogeneities in the barrier height [66]. The inset of Figure 5 shows the linear relationship, and by extrapolating to  $n=1$ , a lateral homogeneous barrier height of 0.54 eV was obtained for SBD before and after irradiation. Patch density can also be used to explain the rise in ideality factor and drop in  $SBH_{IV}$  at lower temperatures [42, 47, 67, 68]. Both the  $IV$  SBH and the ideality factor depend on patch density [69]. When the patch density is larger, it is expected that the ideality factor becomes larger [42, 47]. However, it was observed from the findings that the patch (local region) has smaller barrier heights than the lateral homogenous contacts, which explained the reduction in  $SBH_{IV}$  as the ideality factors increase [42, 47, 70]. These patches allow electrons to preferentially flow through lower barrier areas, distorting  $IV$  characteristics.

To further investigate the anomalous behaviour (that is, an increase in zero-bias SBH with increasing temperature) that occurred in experimental results obtained from  $IV$  characteristics, the approach by Werner and Guttler [71, 72] will be adopted. They suggest that the deviation from ideal behaviours can be explained using the spatial distribution of the SBH at the M-S junction of the SBD. The abnormal behaviours can be described by assuming a Gaussian distribution,  $P(\Phi_B^{IV})$ , of the barrier heights with mean Schottky barrier heights (SBH),  $\overline{\Phi_{B0}^{IV}}$ , and a standard deviation,  $\sigma_s$ , given by [72]

$$P(\Phi_B^{IV}) = \frac{1}{\sigma_s \sqrt{2\pi}} \exp \left[ -\frac{(\Phi_B^{IV} - \overline{\Phi_{B0}^{IV}})^2}{2\sigma_s^2} \right] \quad (7)$$

where the term before the exponential function is the normalisation of the Gaussian barrier height distribution. Incorporating Equation (7) into the TE model yields temperature-dependent expressions for the experimental apparent SBH,  $\Phi_{ap}$  [73]

$$\Phi_{ap} = \overline{\Phi_{B0}^{IV}}(T=0) - \frac{q\sigma_s^2}{2kT} \quad (8)$$

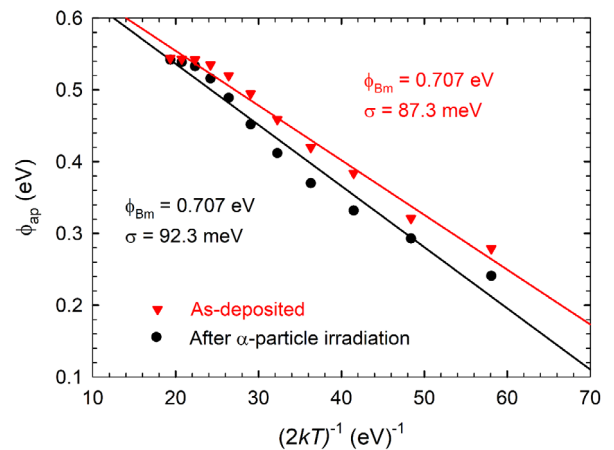
where  $\overline{\Phi_{B0}^{IV}}(T=0)$  is the mean zero bias SBH, which can be determined from the intercept of Figure 6,  $\sigma_{s0}$  is the zero bias standard deviation, which can be evaluated from the slope of Figure 6, and it is generally small and may be neglected.

From Figure 6, the extracted values  $\overline{\Phi_{B0}^{IV}}(T=0)$  were  $\sim 0.707$  eV for both before and after alpha-particle irradiation, while the standard deviations,  $\sigma_{s0}$  were 87.3 meV (before irradiation) and 92.3 meV (after irradiation). The values of  $\sigma_{s0}$  is a measure of the homogeneity of the SBH at the junction of the Cr/n-Si SBD. For homogeneous SBH to occur, the value of  $\sigma_{s0}$  would be zero or very small to the extent that it can be neglected [53]. From the experimental results for  $\sigma_{s0}$ , the values were not actually small when compared to the values of  $\overline{\Phi_{B0}^{IV}}(T=0) = 0.71$  eV. These results confirm the presence of significant barrier inhomogeneities in the Cr/n-Si junction, both before and after alpha-particle irradiation. The observed variation in SBH and ideality factor with temperature (especially at low temperatures) is attributed to these inhomogeneities and may also be influenced by structural phase changes due to cooling [45].

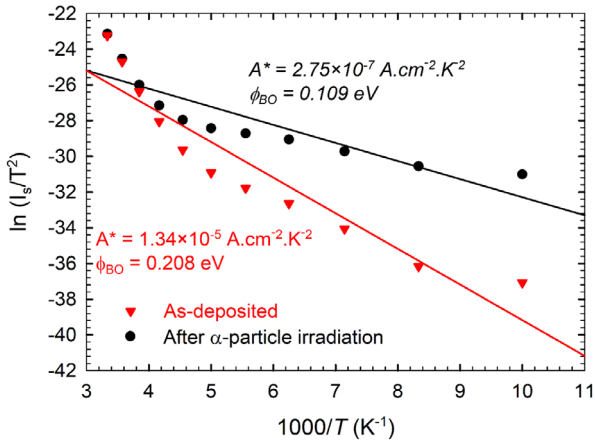
To determine the classical SBH, it is expected to use a conventional TE model. Based on the reasons observed and described earlier, there was a deviation from the classical TE model. Therefore, the Richardson constant and mean barrier height can be determined by rewriting the saturation current in Equation (5) as

$$\ln \left( \frac{I_s}{T^2} \right) = \ln(AA^*) - \frac{q\Phi_{B0}^{IV}}{kT} \quad (9)$$

Figure 7 depicts the conventional Richardson plots of  $\ln(I_s/T^2)$  as a function of  $1000/T$  for Cr/n-Si SBD before and after irradiation, which deviated from linearity at low temperatures. According to the thermionic model, the slope of the plot should give the effective SBH, and the intercept gives the conventional Richardson constant. Before and after irradiation of the Cr/n-Si SBD, the experimental values of the effective Richardson constant obtained were  $1.3 \times 10^{-5}$  and  $2.8 \times 10^{-7} \text{ A.cm}^{-2} \cdot \text{K}^{-2}$ , respectively. The



**FIGURE 6** | A plot of the  $IV$  Schottky barrier height versus inverse of temperature  $(2kT)^{-1}$  for Cr/n-Si Schottky barrier contact to determine the mean Schottky barrier height and its standard deviation both before and after being irradiated by alpha-particles.



**FIGURE 7** | The Richardson plots for the Cr/n-Si SBD before and after irradiation by alpha-particles at a fluence of  $1.3 \times 10^{10} \text{ cm}^{-2}$ .

experimental values obtained were extremely small when compared to the theoretical value of the Richardson constant for electrons in n-Si (i.e.,  $112 \text{ A.cm}^{-2}.\text{K}^{-2}$ ) [53, 58, 60]. The deviation from the value may be related to the lateral inhomogeneities SBHs, crystal defects and potential fluctuation at the junction of Cr and n-Si, which are caused by low and high barrier areas [74]. The plots confirmed that the devices before and after irradiation are highly dependent on temperature, which caused the distortion of the plots from linearity.

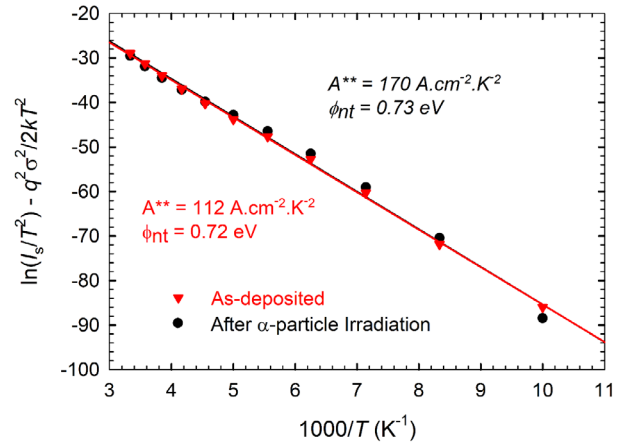
In order to correct the extremely small value of the Richardson constant, the inhomogeneities of the zero-bias SBH must be considered. Therefore, Equation (9) can be rewritten to accommodate the Gaussian distribution of zero-bias SBH [75]

$$\ln \left( \frac{I_s}{T^2} \right) - \left( \frac{q^2 \sigma_{s0}^2}{2k^2 T^2} \right) = \ln (AA^{**}) - \frac{q\overline{\Phi}_{B0}}{kT} \quad (10)$$

From Equation (10), the modified Richardson plots of  $\ln(I_s/T^2) - (q^2 \sigma_{s0}^2 / 2k^2 T^2)$  as a function  $1000/T$  by using the experimental values of saturation current for the SBD before and after irradiation are plotted in Figure 8. The term  $q^2 \sigma_{s0}^2 / 2k^2 T^2$  is the correction factor, taking into account the Gaussian distribution of the barrier heights, and  $A^{**}$  is the modified Richardson constant, which can be calculated from the intercept, and all other parameters have been defined. After the calculation, the mean SBH,  $\overline{\Phi}_{B0}$ , obtained from the slope of the plots were 0.72 and 0.73 eV, before and after irradiation of Cr/Si SBD, respectively, aligning closely with Figure 6's result ( $\sim 0.71 \text{ eV}$ ). The modified Richardson constants,  $A^{**}$ , evaluated from the intercept of Figure 8 are  $112 \text{ A.cm}^{-2}.\text{K}^{-2}$  (matching the theoretical value for n-Si) before irradiation [53, 58, 60] and  $170 \text{ A.cm}^{-2}.\text{K}^{-2}$  after. The deviation from the theoretical value after irradiation may be attributed to the presence of new deep-level defects induced.

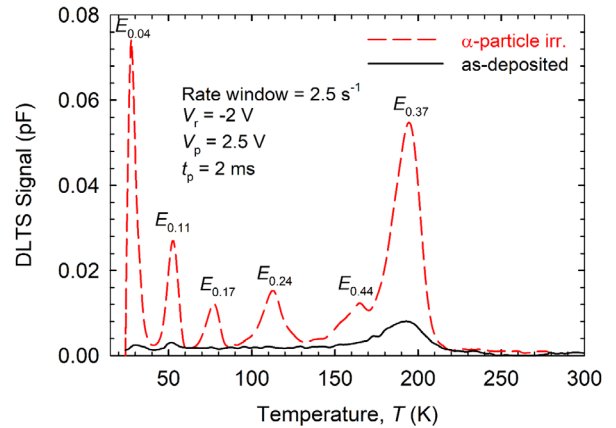
### 3.4 | Deep Level Transient Spectroscopy (DLTS)

To fully understand the variations observed in the values of the electrical IV and CV parameters obtained in the Cr/n-Si Schottky diodes after alpha-particle irradiation, it is essential to investigate the presence of electrically active defects. It has been

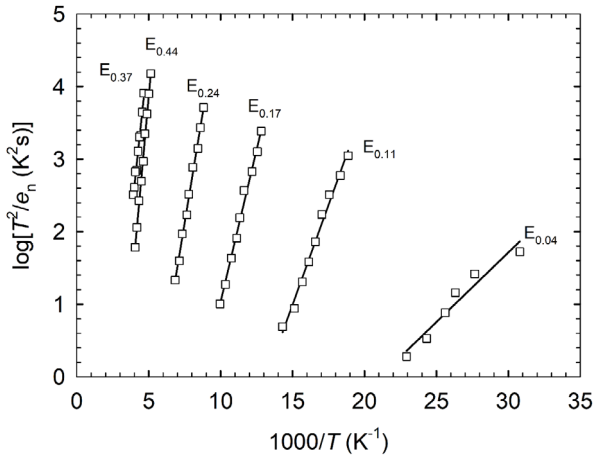


**FIGURE 8** | The modified Richardson plots for the Cr/n-Si SBD before and after irradiation by alpha-particles at a fluence of  $1.3 \times 10^{10} \text{ cm}^{-2}$ .

reported that resistive evaporation doesn't introduce measurable defects [36]. However, previous studies have shown that EBD induces electrically active defects into devices [36, 76]. DLTS is a powerful and non-destructive technique to characterise deep-level defects present in devices by determining their signature at emission rate window of  $2.5$  to  $1000 \text{ s}^{-1}$  [77]. Figure 9 depicts the DLTS spectra of the Cr/Si SBDs before and after alpha-particle irradiation, recorded over the temperature range of 24 to 300 K at a rate window of  $2.5 \text{ s}^{-1}$ , with a quiescent reverse bias of 2.0 V, filling pulse of 2.5 V and filling pulse width of 2.5 ms. Prior to irradiation, three small peaks were observed, while six prominent peaks were observed after irradiation. Notably, three of the post-irradiation peaks aligned with those present in the as-deposited state, which suggests enhancement of pre-existing defects and the formation of new ones due to alpha-particle irradiation. The signatures (activation energy  $E_n$ , where  $E$  is the electron trap, and  $n$  after  $E$  stands for activation energy, and apparent capture cross section,  $\sigma_n$ ) of the deep level defects induced were determined from the slope and the intercept of an Arrhenius plot of  $\log[T^2/e_n]$  as a function of  $1/T$ , according to [77]



**FIGURE 9** | The conventional DLTS spectrum of deep level defects induced in n-Si after irradiation by alpha-particles at a fluence of  $1.3 \times 10^{10} \text{ cm}^{-2}$  recorded at a rate window of  $2.5 \text{ s}^{-1}$ .



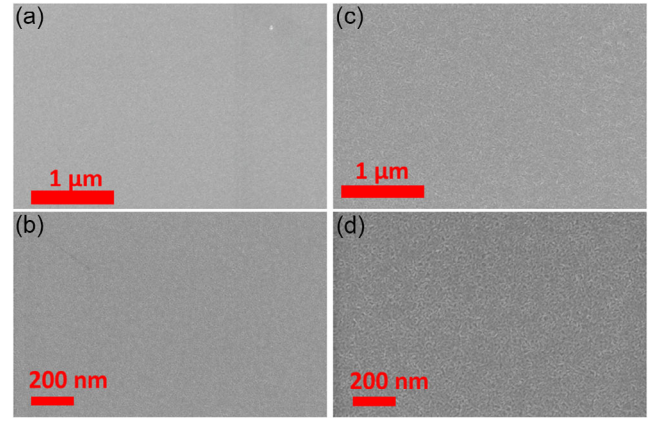
**FIGURE 10** | The Arrhenius plots of deep level defects introduced in n-Si after irradiation by alpha-particles at a fluence of  $1.3 \times 10^{10} \text{ cm}^{-2}$  determined at various rate windows (2.5 to  $1000 \text{ s}^{-1}$ ).

$$e_n(T) = \sigma_n \langle v_n \rangle \frac{g_0}{g_1} N_C \exp \left[ -\frac{E_C - E_T}{kT} \right] \quad (11)$$

where  $e_n(T)$  is the emission rate as a function of temperature (2.5 to  $1000 \text{ s}^{-1}$  was used in this study),  $\langle v_n \rangle$  is the thermal speed of the electron,  $N_C$  is the density of states in the conduction band,  $E_C - E_T = E_n$  is the activation energy, and  $g_0$  and  $g_1$  are degeneracy terms of the defect states before and after electron emission. Figure 10 shows the Arrhenius plots used to extract the activation energies and apparent capture cross sections of each deep level defect from the slopes and intercepts of the linear fits. The three intrinsic deep levels detected before irradiation were labelled  $E_{0.04}$ ,  $E_{0.11}$  and  $E_{0.37}$ , and their electronic properties are tabulated in Table 3. After irradiation with alpha-particles, three additional electrically active defects labelled  $E_{0.17}$ ,  $E_{0.24}$  and  $E_{0.44}$  emerged, and the amplitudes of the pre-existing traps increased significantly. The activation energies, apparent capture cross sections and attributions of the defects observed are tabulated in Table 3. The appearance of new deep levels and the enhancement of the existing ones directly correlate with the degradation in diode characteristics, which affirms that the changes in *IV* and *CV* behaviour after irradiation result from the introduction of additional trap states.

**TABLE 3** | Electronic properties of deep level defects induced in Cr/n-Si after irradiation by alpha-particles at a fluence of  $1.3 \times 10^{10} \text{ cm}^{-2}$ .  $T_{2.5}$  stands for the peak temperature at a rate window of  $2.5 \text{ s}^{-1}$ .

| Defect label | $E_n$ , eV | $\sigma_n$ , $\text{cm}^2$ | $T_{2.5}$ K | Defect ID             |
|--------------|------------|----------------------------|-------------|-----------------------|
| $E_{0.04}$   | 0.04       | $3.2 \times 10^{-18}$      | 28          | —                     |
| $E_{0.11}$   | 0.11       | $5.1 \times 10^{-15}$      | 53          | $C_i$ [78, 79]        |
| $E_{0.17}$   | 0.17       | $6.3 \times 10^{-15}$      | 77          | $VO, C_i C_s$ [78–80] |
| $E_{0.24}$   | 0.24       | $2.5 \times 10^{-15}$      | 113         | $V_2^{=/-}$ [80]      |
| $E_{0.44}$   | 0.44       | $4.4 \times 10^{-15}$      | 166         | $V_2^{-/0}$ [21, 81]  |
| $E_{0.37}$   | 0.37       | $2.1 \times 10^{-17}$      | 194         | CP [36, 82]           |



**FIGURE 11** | Scanning electron microscopy at 2.0 kV of the Cr thin films evaporated on n-Si. The images (a) and (b) were recorded from the as-deposited samples at magnifications of 20 and 50 kX, respectively; and (c) and (d) were recorded after irradiating the sample with alpha-particles at a fluence of  $1.3 \times 10^{10} \text{ cm}^{-2}$  at magnifications of 20 and 50 kX, respectively.

### 3.5 | Morphological Studies

Scanning electron microscopy (SEM) was employed to investigate the surface morphology of Cr/n-Si film before and after alpha-particle irradiation in order to correlate the morphological characteristics with the electrical behaviour of the devices. Figure 11a–d shows plane-view SEM micrographs captured at an accelerating voltage of 2.0 kV under the magnifications of 20 000 and 50 000. Figure 11a,b show the images of as-deposited Cr/n-Si surfaces. At lower magnification (20 000), the chromium film appeared fine, continuous, and homogeneously deposited on the n-Si substrate. At higher magnification (50 000), localised surface inhomogeneities (seen as patches) were clearly revealed. These patches, commonly observed of EBD, are likely sources of the Schottky barrier height inhomogeneities at the Cr/n-Si interface. Additionally, small void-like features were observed at closer inspection of the image at higher magnification, which may contribute to spatial variations in current transport and impact device performance. Figure 11c,d depict the surface after irradiation at magnifications of 20 000 and 50 000, respectively. While the general continuity of the Cr film remained, slight changes such as enhanced surface roughness and increased patchiness were evident, particularly at 50 000. These alterations at the surface could result from atomic displacement effects induced by alpha-particle irradiation, which potentially increases the density of inhomogeneities. The observed increase in surface inhomogeneities and patch density after irradiation supports the earlier electrical and DLTS findings, which suggest that these morphological characteristics may contribute to enhanced barrier inhomogeneity and deep-level trap formation at the interface of the Cr/n-Si [47, 83].

### 4 | Conclusions

In this study, Cr/n-Si SBDs fabricated by EBD were comprehensively characterised from 60 to 300 K before and after 5.4 MeV alpha-particle irradiation from a  $^{241}\text{Am}$  source at a fluence of



$1.3 \times 10^{10} \text{ cm}^{-2}$ . The diodes exhibited excellent rectification in as-deposited state, with an ideality factor of 1.04 and SBH of 0.54 eV (IV) and 0.67 eV (CV). Temperature-dependent CIV and CV analyses revealed non-ideal behaviour, particularly at cryogenic temperatures, where the ideality factor increased and the SBH decreased. These trends were further accentuated after irradiation which indicate enhanced barrier inhomogeneity due to radiation-induced defects. The extracted Richardson constants deviated significantly from the theoretical value for n-type Si ( $112 \text{ A.cm}^{-2}.\text{K}^{-2}$ ) when ideal thermionic emission was assumed. However, after accounting for barrier inhomogeneity via a Gaussian distribution model, the modified Richardson constants closely matched the theoretical value before irradiation ( $112 \text{ A.cm}^{-2}.\text{K}^{-2}$ ) and showed only a modest increase after irradiation ( $170 \text{ A.cm}^{-2}.\text{K}^{-2}$ ). Corresponding mean barrier heights were  $\sim 0.72\text{--}0.73 \text{ eV}$ , which was consistent with expectations for Cr/n-Si contacts. DLTS confirmed the introduction of new electrically active defects after irradiation, alongside the enhancement of pre-existing intrinsic traps. These defects correlate directly with the observed changes in carrier transport and barrier uniformity. SEM further supported these findings, showing increased surface patchiness and roughness after irradiation, consistent with the electrically inferred inhomogeneities. Despite the introduction of defects and increased interfacial disorder, Cr/n-Si SBDs maintained stable rectifying characteristics across the investigated temperature range, even after irradiation. This resilience demonstrates the potential of Cr-based Schottky contacts for use in radiation-prone or cryogenic environments, where both temperature stability and radiation hardness are required.

## Acknowledgments

Financial support from the South Africa NRF (Grant no. 120856), as well as from the University of Johannesburg (UJ) URC and FRC (Grant nos. 282810, 083135), is acknowledged by A.R.E.P. and C.J.S. E.O. acknowledges TETFUND for the Conference Attendance Intervention.

## Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

## Data Availability Statement

The datasets generated during and/or analysed during the current study are available from the corresponding author upon reasonable request.

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